

Aufgabe 3:

(a) $\frac{\text{DWA} \quad \text{--- (keine Prämisse)}}{\text{(do while)} \quad \langle \text{do } S \text{ while } b, \varnothing \rangle \Rightarrow \langle S; \text{while } b \text{ do } S, \varnothing \rangle}$

for $\frac{\text{---}}{\langle \text{for } I \text{ test } b \text{ update } U \text{ do } S, \varnothing \rangle \Rightarrow \langle I; \text{while } b \text{ do } (S; U), \varnothing \rangle}$

(b) (SWK 4 Seite 41)

$\langle i := 42; \text{do } (\text{for } j := 19 \text{ test } j > 0 \text{ update } j := j - 20 \text{ do } i := i + j) \text{ while } i < 60, \{ \} \rangle \xRightarrow{\text{Begründung}} \langle \text{do } A \text{ while } i < 60, \{ i \mapsto 42 \} \rangle$

$\textcircled{*}: \frac{\text{---} \quad \text{Ass}}{\langle i < 42, \{ \} \rangle \Rightarrow \{ i \mapsto 42 \}}$

$\langle \underbrace{i := 42}_{S_1}; \text{do } \underbrace{A}_{S_2} \text{ while } i < 60, \{ \} \rangle \Rightarrow \langle \text{do } A \text{ while } i < 60, \{ i \mapsto 42 \} \rangle \quad \text{Comp}_2$

$\xRightarrow{\text{**}} \langle A; \text{while } i < 60 \text{ do } A, \{ i \mapsto 42 \} \rangle$

$\xRightarrow{\text{**}} \langle j := 19; \text{while } j > 0 \text{ do } (i := i + j; j := j - 20); \text{while } i < 60 \text{ do } A, \{ i \mapsto 42 \} \rangle$

$\xRightarrow{\text{Comp}_2, \text{Ass}} \langle \text{while } j > 0 \text{ do } (\dots); \underbrace{\text{while } \dots, \{ i \mapsto 42, j \mapsto 19 \}}_B \rangle$

$\xRightarrow{\text{Comp}_1, \text{WAT?}} \dots \Rightarrow \{ i \mapsto 61, j \mapsto -1 \} \quad \text{--- } S_1 \quad S_2$

$\textcircled{**}: \frac{\text{---}}{\langle \text{do } A \text{ while } i < 60, \{ i \mapsto 42 \} \rangle \Rightarrow \langle A; \text{while } i < 60 \text{ do } A, \{ i \mapsto 42 \} \rangle} \quad \text{DWA}$

$\textcircled{**}: \frac{\langle \text{for } j := 19 \text{ test } j > 0 \text{ update } j := j - 20 \text{ do } i := i + j, \{ i \mapsto 42 \} \rangle \Rightarrow \langle j := 19; \text{while } j > 0 \text{ do } \underbrace{i := i + j}_{S_1'}; \underbrace{j := j - 20}_{S_2'} \rangle}{\langle \underbrace{A}_{S_1}; \underbrace{B}_{S_2}, \{ i \mapsto 42 \} \rangle \Rightarrow \langle j := 19; \text{while } j > 0 \text{ do } (i := i + j; j := j - 20); \underbrace{B}_{S_2'} \{ i \mapsto 42 \} \rangle} \quad \text{For, Comp}_1$

$\therefore \frac{\text{---} \quad \text{WAT } \llbracket b \rrbracket(b) = 1}{S_1 \Rightarrow S_1'} \quad \text{Comp}_1$
 $S_1; S_2, \varnothing \Rightarrow S_1'; S_2, \varnothing$